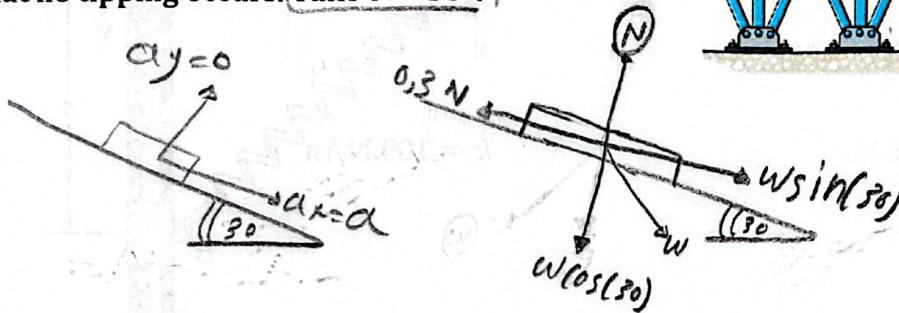
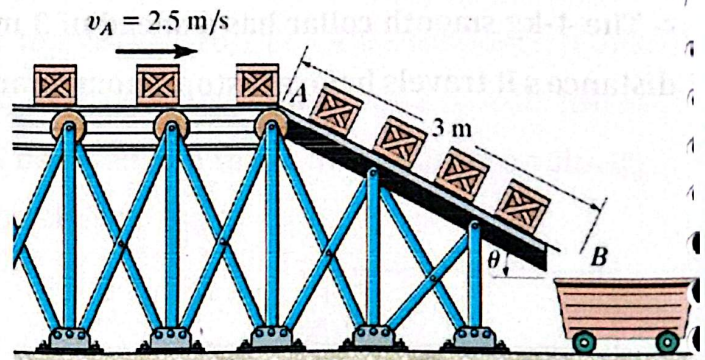


Name:

Section No.:

SHEET # 2 (Rectilinear Motion of a Particle)

- 1) The conveyor belt delivers each 12-kg crate to the ramp at A such that the crate's speed is $v_A = 2.5 \text{ m/s}$, directed down along the ramp. If the coefficient of kinetic friction between each crate and the ramp is $\mu_k = 0.3$, determine the speed at which each crate slides off the ramp at B. Assume that no tipping occurs. Take $\theta = 30^\circ$.



$$\begin{aligned} V_0 &= 2.5 \text{ m/s} \\ V_f &= \dots ?? \\ a &= \text{constant} \\ S &= 3 \text{ m} \end{aligned}$$

$$\sum F(x) = ma$$

$$W \sin(30) - 0.3N = m(a_x)$$

$$\Rightarrow m g \sin(30) - 0.3(m g \cos(30)) = m a_x$$

$$\sum F(y) = 0 \mid g \sin(30) - 0.3(g \cos(30)) = a_x = 2.35 \text{ m/s}^2$$

$$N = W \cos(30)$$

$$V_f^2 = V_0^2 + 2a_x S$$

$$V_f = \sqrt{(2.5)^2 + 2(2.35)(3)}$$

$$\therefore V_B = 4.5 \text{ m/s}$$

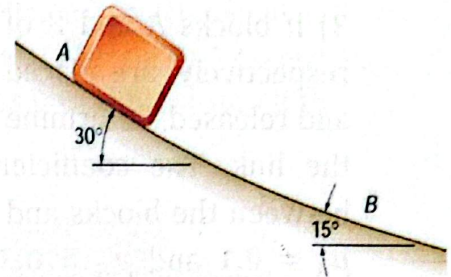
Chapter Two: Kinetics of Particles: Force and Acceleration

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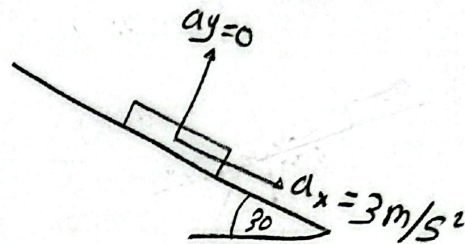
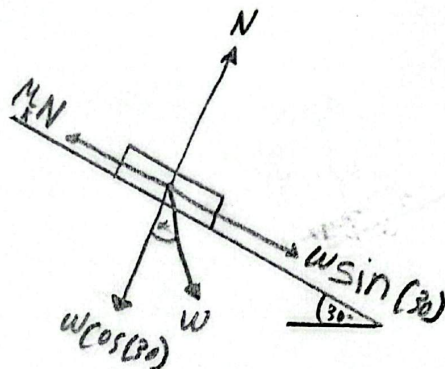
Section No.:

SHEET # 2 (Rectilinear Motion of a Particle)

- 2) The acceleration of a package sliding at point A is 3 m/s^2 . Assuming that the coefficient of kinetic friction is the same for each section, determine the acceleration of the package at point B.



[A]



$$\sum F_x = ma$$

$$W \sin(30) - \mu_k N = m(3) \Rightarrow$$

$$\sum F_y = 0 \Rightarrow N \cos(30) - W \cos(30) = 0 \Rightarrow N = W$$

$$\therefore g \sin(30) - \mu_k = 3 \Rightarrow \mu_k = 0.224$$

$$N = m(9.81) \cos(30)$$

$$\sum F_x = ma$$

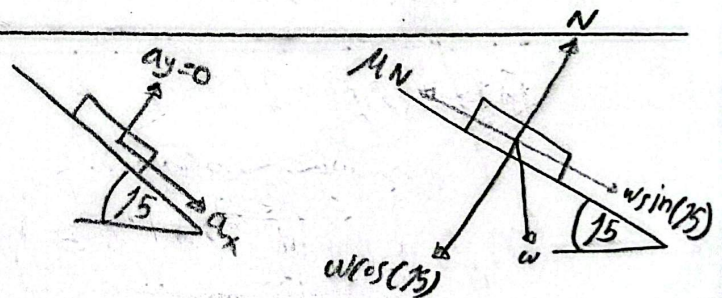
$$\sum F_y = 0$$

$$W \sin(15) - 0.22 N = m a_x$$

$$N = W \cos(15)$$

$$g \sin(15) - 0.22 g \cos(15) = a_x$$

$$g \sin(15) - 0.22(9.81) \cos(15) = a_x \Rightarrow a_x = 0.45 \text{ m/s}^2$$



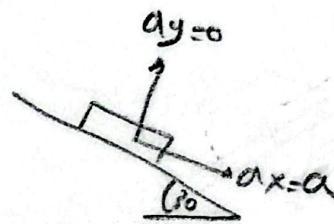
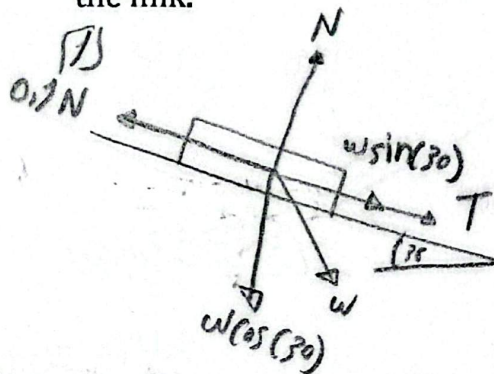
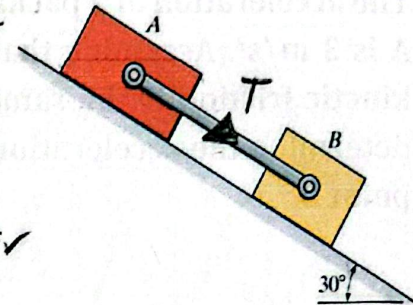
Chapter Two: Kinetics of Particles: Force and Acceleration

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SHEET # 2 (Rectilinear Motion of a Particle)

3) If blocks A and B of mass 10^A kg and 6^B kg respectively, are placed on the inclined plane and released, determine the force developed in the link. The coefficients of kinetic friction between the blocks and the inclined plane are $\mu_A = 0.1$ and $\mu_B = 0.3$. Neglect the mass of the link.



$$\sum F_x = ma_x$$

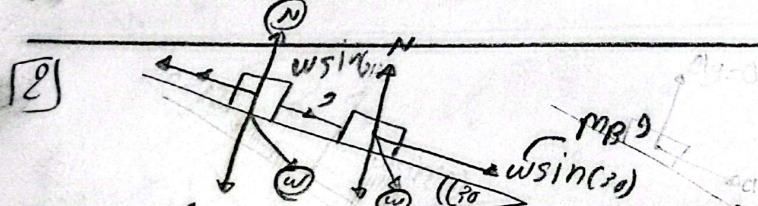
$$W \cos(30)$$

$$T + W \sin(30) - 0.1N = ma_x$$

$$T_{AB} + [(10)(9.81)] \sin(30) - 0.1[(10)(9.81)] (\cos(30)) = 10 \cdot a_x$$

$$\sum F_y = ma_y \quad \therefore T_{AB} = -6.35 = 6.35 \text{ "C"}$$

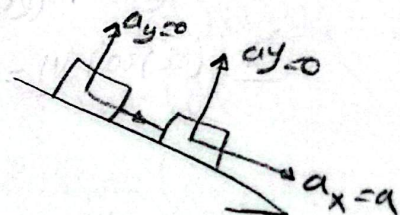
$$N = W \cos(30)$$



$$\sum F_x = (m_B g) \sin(30) + (m_A g) \sin(30) - N_A \mu_A - N_B \mu_B = (m_A + m_B) a_x$$

$$\sum F_y = N_A = m_A g \cos(30); N_B = m_B g \cos(30)$$

$$a = 3.42 \text{ m/s}^2$$



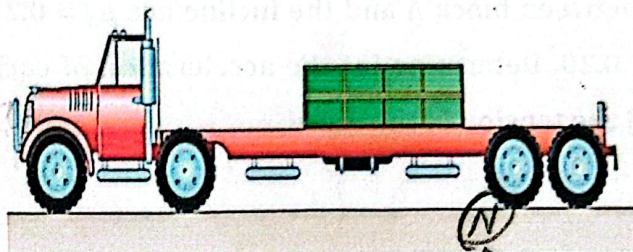
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SHEET # 2 (Rectilinear Motion of a Particle)

- 4) The coefficients of friction between the load and the flatbed trailer shown are $\mu_s = 0.40$ and $\mu_k = 0.30$. Knowing that the speed of the rig is 72 km/h , determine the shortest distance in which the rig can be brought to a stop if the load is not to shift.



$$\sum F_x = -0.4N = ma$$

$$\sum F_y = -W + N = 0$$

$$N = m \cdot g = W$$

$$-0.4N = m \cdot g \cdot a$$

$$-0.4N = N \cdot a$$

$$a = -3.92$$

$$-0.4mg = ma$$

$$a = -0.4 \cdot 9.81 =$$

* معادلات الحركة المستقيمة

$$V_f^2 = V_o^2 + 2aX$$

$$0 = (20)^2 + 2(-3.92)(X)$$

$$\therefore X = 50.96$$

$$V = 72 \times \frac{1000}{60 \times 60}$$

$$V = 20 \text{ m/s}$$

$$V_f = 0 \text{ m/s}$$

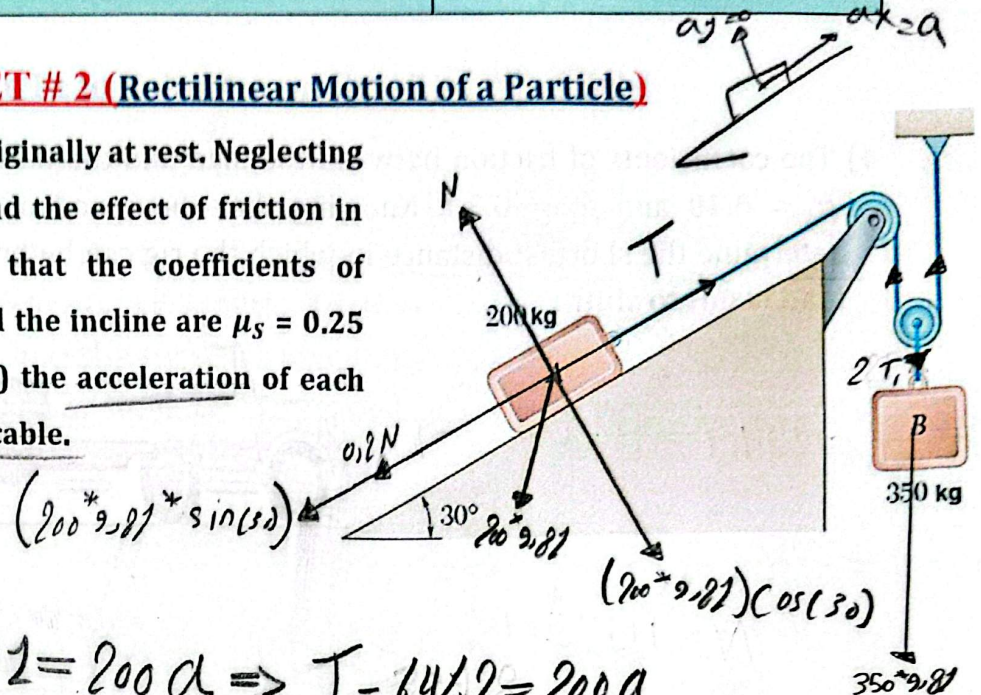
Chapter Two: Kinetics of Particles: Force and Acceleration

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SHEET # 2 (Rectilinear Motion of a Particle)

- 5) The two blocks shown are originally at rest. Neglecting the masses of the pulleys and the effect of friction in the pulleys and assuming that the coefficients of friction between block A and the incline are $\mu_s = 0.25$ and $\mu_k = 0.20$. Determine (a) the acceleration of each block, (b) the tension in the cable.



[A]

$$\sum f_x = T - 0.2N - 981 = 200a \Rightarrow T - 641.2 = 200a$$

$$\sum f_y = N = 1699.1 \text{ N}$$

[B]

$$T_2 = 2T_1$$

$$a_A = 2a_B$$

$$\sum f_x = 0$$

$$\sum f_y = ma_y = 2T - (350 \times 9.81) = 350a_B$$

$$T - 641.2 = 400a_B \Rightarrow T + 400a_B = 641.2$$

$$2T - 3433.5 = 350a_B \Rightarrow 2T + 350a_B = 3433.5$$

$$T = 2553.3 \text{ N}$$

$$a_B = 4.8 \text{ m/s}^2$$

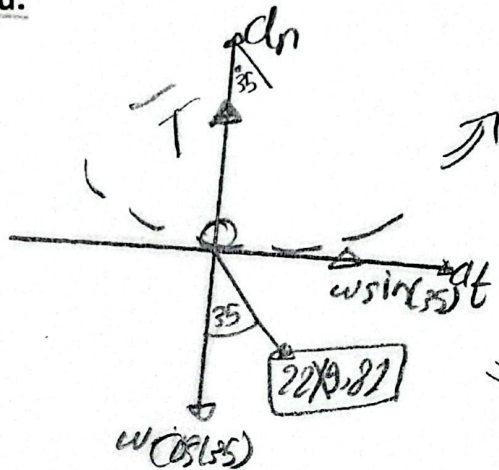
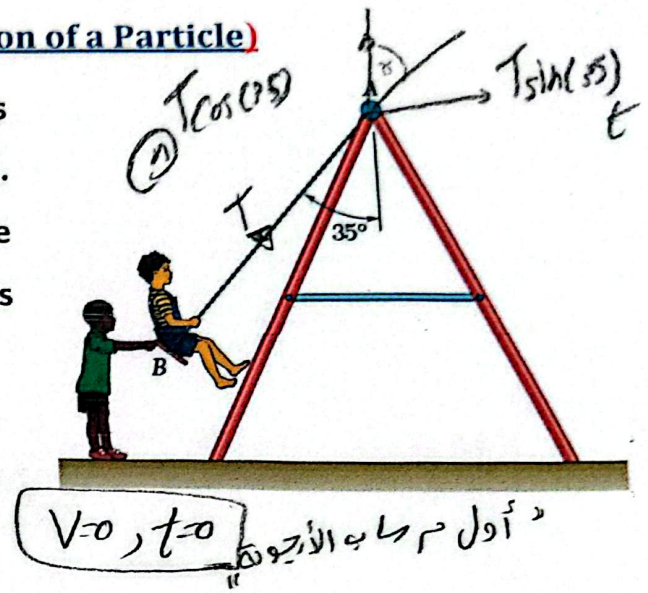
Chapter Two: Kinetics of Particles: Force and Acceleration

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SHEET # 2 (Rectilinear Motion of a Particle)

6) A child having a mass of 22 kg sits on a swing and is held in the position shown by a second child. Neglecting the mass of the swing, determine the tension in rope AB immediately after the swing is released.



$$\Rightarrow \sum F_{(t)} = W \sin(35) = 22 \, dt$$

$$\sum F_{(n)} = m a_n$$

$$T - mg \cos(35) = m \frac{v^2}{r} = 0$$

$$\boxed{T = mg \cos(35)}$$